MultiProbe Pico Debugger

General Description

A debug probe for the RP2040 & RP2350 devices that can debug up to 4 devices at the same time with two active UARTs.

Key features

- Debug multiple picos using a single debugger over a single USB cable.
- Connect to each uart from a single USB (2 active limit at the moment).
- Indicator LED for active SWD probes and activated UARTS.

Required software

- Linux operating system
- VSCode with the Raspberry Pi Pico extension

Software documentation

By default DEBUG 1 should work out the box like the official debugprobe. To target the other debug ports, an openOCD command must be set eg. "cmsis-dap usb interface 1" for DEBUG 2. For the latest documentation and UART setup please see the github project readme (https://github.com/picolemon/multiprobe).

Applications

- Device debugging

Pinout

DEBUG 1-4 SWD Connector

Pin	Pin name	Description
1	SWDIO	SWD data
2	GND	Ground
3	SWDCLK	SWD clock

UART 1-4 Connector

Pin	Pin name	Description
1	GND	Ground
2	TX	UART Transmit pin (out)
3	RX	UART Receive pin (in)

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PWROUT Connector

Pin	Pin name	Description
1	5V	USB VBUS 5V output
2	GND	Ground
3	RX	3V power output of the ME6217C33M5G DC to DC converter (Low power output, don't use to power any external device)

Host interface

USB-C compatible

Buttons

Btn	Description
BOOT	Flash device, hold down during powerup or post reset to run the UF2.Bootloader
RESET	Hardware reset device

SWD Header (Unpopulated)

Unpolulated SWD interface to debug the onboard RP2350, used for debugger development so ignore for general use.

Pin	Pin name	Description
1	SWDIO	SWD data
2	GND	Ground
3	SWDCLK	SWD clock

UART Header (Unpopulated)

Unpolulated UART interface to debug the onboard RP2350, used for debugger development so ignore for general use.

Pin	Pin name	Description
1	TX	UART Transmit pin (out)
2	GND	Ground
3	RX	UART Receive pin (in)

Links

Github: https://github.com/picolemon/multiprobe

Change history

01-10-25: Initial version.

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Contact & support

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